

Model Refinement Report

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1 Introduction

This report is a continuation of the previous paper discussing the quality of the baseline and the LSTM models. In particular, I highlight the changes made to increase the quality of the forecasts.

2 Method

As in the previous paper, the evaluation of the models is performed on the US futures gold prices data set with the hourly rate of the records. The data set captures prices from Summer 2023 and ends in the beginning of July 2025.

The test set is set to be the beginning of July 2025 with the previous data being the train set.

Furthermore, the data set frequency was fixed to 30 minutes. The missing data was filled backward and forward.

The following metrics were used: mean squared error, mean absolute error, and the mean absolute percentage error. The first metric captures significant errors of the predicted values, while the second metric identifies the absolute error values. Finally, the last metric can be used to evaluate accuracy of a model.

In addition, the predicted and true prices on the test set are plotted.

2.1 Savitzky-Golay Filter

As proposed by the Wang et al. [1], the Savitzky-Golay filter was applied for the close, high, low, and open prices as well as volume time series.

2.2 Baseline Model

The following ensemble architecture is used for the baseline model:

1. High price predictor (ARIMA)
2. Low price predictor (ARIMA)

3. Volume predictor (XGBoost regressor)
4. Close price predictor (XGBoost regressor)

2.3 LSTM Model

The following architecture of the LSTM model is proposed:

```
model = Sequential([
    Input(shape=(19, 60)),
    LSTM(128, return_sequences=True),
    Dropout(0.3),
    LSTM(64),
    Dense(32, activation='relu'),
    Dense(len(self.feature_names))
])
model.compile(optimizer='adam', loss='mse')
```

The following changes were made:

1. Change target (output) features to close, high, low prices, and volume.
2. Temporal features were added

3 Results

The results of the comparison include the following metrics:

3.1 Metrics

Model	MSE	MAE	MAPE
Baseline	265	13	0.04
LSTM	521	21	0.006

Table 1: Metrics results

3.2 Plots

The predictions of prices compared with the true values are shown below in the figures [Fig. 1](#) and [Fig. 2](#). The blue curve represents the true values of the gold prices, while the orange line shows the predictions of each model.

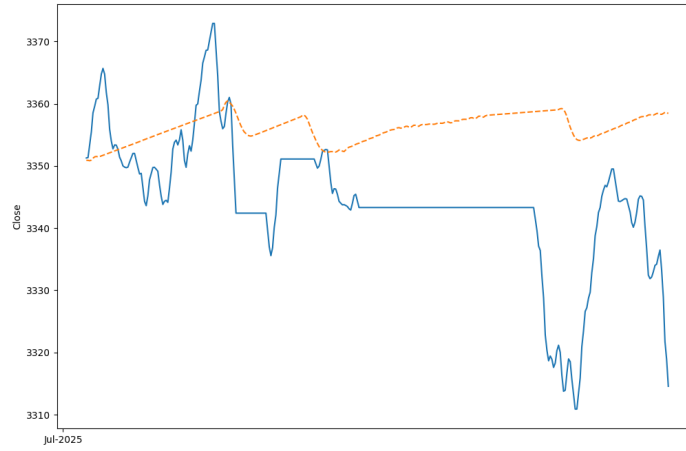


Figure 1: Baseline model predictions vs. true prices

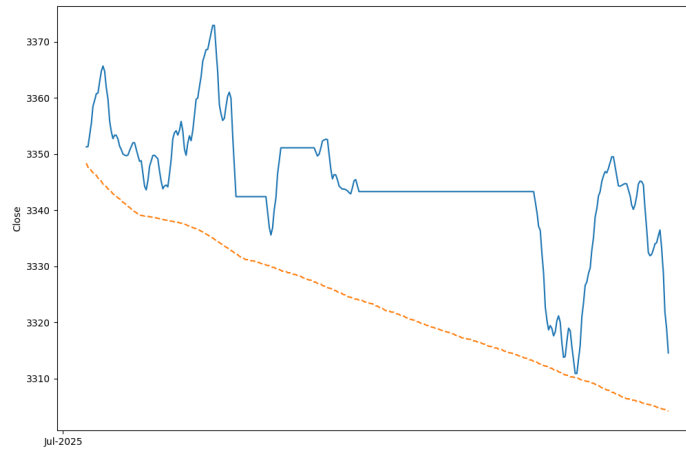


Figure 2: LSTM model predictions vs. true prices

4 Discussion

The adjustments in the models helped approximate their predictive power. The baseline model demonstrates non-linear behavior and follows some dynamics of

the real time series. However, the model lacks precision in the amount of drops and rises.

The LSTM model has smaller bias now compared with the previous version, though the predictions are still valid only within a small period of time.

5 Conclusion

In conclusion, the baseline model still gets closer to the true values in general compared to the LSTM model. However, the LSTM model has better MAPE metric suggesting better accuracy.

References

- [1] Wang et al., “Two-stage stock portfolio optimization based on AI-powered price prediction and mean-CVaR models,” *Expert Syst. Appl.*, vol. 255, 2024. [Online] Available: <https://www.sciencedirect.com/science/article/pii/S0957417424014222>